



# Hardware Test Report

Electronics Projects

## Abstract

This document serves the purpose of planning, testing and documenting the developed hardware in the project module 3.

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# 1. Test procedure

After the hardware has been completely soldered and assembled, it is of utmost importance to carefully inspect the solder joints under a microscope. This pre-testing step can help prevent potential damage to the electronics caused by shorts or misplaced components.

Next, the “life signs” test can begin, where the board is powered on for the first time. During this step, all safety considerations mentioned in the safety chapter must be observed. All internal voltages and current draws should be documented and compared against the expected values. In case of a failure, a structured debugging approach must be followed. If the life signs test passes, it is permissible to proceed to the basic hardware functionality tests.

The basic hardware functionality test aims to verify the operation of sub-circuits groups. It follows a systematic approach, starting from the audio input connectors, through the analog input circuitry, codec, analog output circuitry, and ending at the output connectors.

After verifying the basic hardware functionality, an additional test will be performed to determine the corner frequency of the board’s input/output filter. A frequency sweep across the human-audible spectrum will be applied to the board’s input, and a Fourier transform of the output signal will be used to generate an amplitude response plot.

# 2. Safety

Although not part of the testing, care must be taken when soldering and handling the soldering iron. Soldering temperatures range from 300 °C to 400 °C and can cause serious burns if not handled properly.

When powering the board for the first time, it is good practice to use a laboratory power supply with a current-limit feature to prevent severe damage in case of a short circuit. Even better results can be achieved by powering the board at a slightly lower voltage initially to further limit power in case of a fault.

If abnormal behavior such as smoke, overheating, or excessive current draw is observed, disconnect power immediately and inspect the hardware before continuing.

Work on a ESD protected surface to prevent ESD damage to components.

### 3. Test Plan

| Test item, Description                                                                                                                                    | Test conditions                                     | Expected, Tolerance                                                                                                         | Measured            | Comparison, verdict                                               | Remarks, Discussion                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| General setup for all the following tests (unless stated otherwise):<br>- Power in by laboratory power supply, 5V<br>- +3V0 (P2, Pin 1) connected to +3V3 |                                                     |                                                                                                                             |                     |                                                                   |                                                                                                             |
| Voltage at +3V3 ESP supply                                                                                                                                | No input signal                                     | $3.32V \mid 33.2k\Omega (\pm 1\%) \cdot 100\mu A (\pm 1\%) = 3.25V \dots 3.39V$                                             | 3.29V               | In range → <b>passed</b>                                          |                                                                                                             |
| Voltage at +3V3 digital supply (VD)                                                                                                                       | No input signal                                     | $3.32V \mid 33.2k\Omega (\pm 1\%) \cdot 100\mu A (\pm 1\%) = 3.25V \dots 3.39V$                                             | 3.29V               | In range → <b>passed</b>                                          |                                                                                                             |
| Voltage at +5V analog supply (VA)                                                                                                                         | No input signal                                     | $4.99V \mid 49.9k\Omega (\pm 1\%) \cdot 100\mu A (\pm 1\%) = 4.89V \dots 5.09V$                                             | 4.99V               | In range → <b>passed</b>                                          |                                                                                                             |
| Voltage at DMX output (DMX_+5V)                                                                                                                           | No input signal                                     | $5.4V \mid 270\Omega (\pm 1\%) \cdot 20mA (\pm 1\%) = 5.29V \dots 5.51V$                                                    | 5.24V               | Out of range, considered acceptable <sup>*)</sup> → <b>passed</b> | *) Taking into account the tolerance envelope                                                               |
| Voltage at Boost output                                                                                                                                   | No input signal                                     | $5.51V \mid \left( \frac{301k\Omega (\pm 1\%)}{30k\Omega (\pm 1\%)} + 1 \right) \cdot 500mV (\pm 15mV) = 5.25V \dots 5.78V$ | 5.67V               | In range → <b>passed</b>                                          |                                                                                                             |
| Ripple on +3V3 ESP supply                                                                                                                                 | No input signal                                     | $0.8\mu V_{RMS} + 1mV \cdot 10^{-\frac{50dB}{20}} \rightarrow < 0.80\mu V$                                                  | 4.0mV <sub>pp</sub> | Out of range, considered acceptable <sup>*)</sup> → <b>passed</b> |                                                                                                             |
| Ripple on +3V3 digital supply (VD)                                                                                                                        | No input signal                                     | $0.8\mu V_{RMS} + 1mV \cdot 10^{-\frac{50dB}{20}} \rightarrow < 0.80\mu V$                                                  | 4.4mV <sub>pp</sub> | Out of range, considered acceptable <sup>*)</sup> → <b>passed</b> | *) Oscilloscope has not enough accuracy, measures above the expected value even when no signal is connected |
| Ripple on +5V analog supply (VA)                                                                                                                          | No input signal                                     | $0.8\mu V_{RMS} + 10mV \cdot 10^{-\frac{50dB}{20}} \rightarrow < 0.81\mu V$                                                 | 1.5mV <sub>pp</sub> | Out of range, considered acceptable <sup>*)</sup> → <b>passed</b> |                                                                                                             |
| Ripple on DMX output (DMX_+5V)                                                                                                                            | No input signal                                     | $< 40mV_{pp}$                                                                                                               | 6mV <sub>pp</sub>   | In range → <b>passed</b>                                          |                                                                                                             |
| Ripple on Boost output                                                                                                                                    | No input signal                                     | $\frac{1A \cdot 0.18}{1MHz \cdot 10\mu F} \rightarrow < 18.3mV$                                                             | 5mV <sub>pp</sub>   | In range → <b>passed</b>                                          |                                                                                                             |
| Current on Power in at 5 V                                                                                                                                | No input signal                                     | $< 180mA$                                                                                                                   | 173mA               | In range → <b>passed</b>                                          |                                                                                                             |
| Current on Power in at 5V, +3V0 disconnected                                                                                                              | No input signal, +3V0 disconnected                  | $< 160mA$                                                                                                                   | 128mA               | In range → <b>passed</b>                                          |                                                                                                             |
| Current ripple on Power in at 5 V                                                                                                                         | No input signal                                     | $\frac{1mV}{5V/180mA} \rightarrow < 36\mu V$                                                                                | 1mA <sub>pp</sub>   | Out of range, considered acceptable <sup>*)</sup> → <b>passed</b> | *) Measurement method provides not enough accuracy                                                          |
| Microphone voltage                                                                                                                                        | No input signal                                     | $2.6V \mid 2V \dots < 5V$                                                                                                   | 2.48V               | In range → <b>passed</b>                                          |                                                                                                             |
| Signal Peak finger snip                                                                                                                                   | No input signal, finger snip next to the microphone | $50mV \mid 0V \dots 2.5V$                                                                                                   | 68mV <sub>pp</sub>  | In range → <b>passed</b>                                          |                                                                                                             |
| Signal Peak finger on microphone                                                                                                                          | No input signal, finger on microphone               | $> 300mV \mid 0V \dots 2.5V$                                                                                                | 500mV <sub>pp</sub> | In range → <b>passed</b>                                          | Note: Expected value given as amplitude, measured peak-to-peak, conversion needed before comparison         |
| Signal Peak finger snip after Preamp                                                                                                                      | No input signal, finger snip next to the microphone | $> 2V \mid 0V \dots 2.5V$                                                                                                   | 5V <sub>pp</sub>    | In range → <b>passed</b>                                          |                                                                                                             |
| Signal Peak finger on microphone after Preamp                                                                                                             | No input signal, finger on microphone               | $> 2V \mid 0V \dots 2.5V$                                                                                                   | 5V <sub>pp</sub>    | In range → <b>passed</b>                                          |                                                                                                             |

|                                                                                                                                                                                                                                                              |                                               |                                                                                                                   |                           |                                                                      |                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Preamp reference voltage (U4)                                                                                                                                                                                                                                | No input signal                               | $2.5V \mid 5V \cdot \frac{100k\Omega \pm 1\%}{(100k\Omega \pm 1\%) + (100k\Omega \pm 1\%)} = 2.475V \dots 2.525V$ | 2.49V                     | In range → <b>passed</b>                                             |                                                                                       |
| Line in signal after Anti-Aliasing filter, left channel                                                                                                                                                                                                      | Input signal:<br>500mVpp 1kHz sine            | No attenuation @ 1kHz → <b>500mV<sub>PP</sub></b>                                                                 | <b>532mV<sub>PP</sub></b> | Acceptable with tolerance → <b>passed</b>                            |                                                                                       |
| Line in signal after Anti-Aliasing filter, right channel                                                                                                                                                                                                     | Input signal:<br>500mVpp 1kHz sine            | No attenuation @ 1kHz → <b>500mV<sub>PP</sub></b>                                                                 | <b>532mV<sub>PP</sub></b> | Acceptable with tolerance → <b>passed</b>                            |                                                                                       |
| Voltage at DMX A output                                                                                                                                                                                                                                      | No input signal                               | $3V + 5V/2 = 5.5V$                                                                                                | <b>4.46V</b>              | Not as expected, considered acceptable <sup>*)</sup> → <b>passed</b> | *) Calculated with max values from the datasheet, is in range for $V_{OD,min} = 1.5V$ |
| Voltage at DMX B output                                                                                                                                                                                                                                      | No input signal                               | $3V - 5V/2 = 0.5V$                                                                                                | <b>0.78V</b>              | Not as expected, considered acceptable <sup>*)</sup> → <b>passed</b> |                                                                                       |
| General setup for all the following tests (unless stated otherwise):<br>- Power in by laboratory power supply, 5V<br>- +3V0 (P2, Pin 1) connected to +3V3<br>- Wire loop SDOUT (P1, Pin 11) to SDIN (P1, Pin 16)<br>- Reset CS4271 Codec after power applied |                                               |                                                                                                                   |                           |                                                                      |                                                                                       |
| Crystal Frequency                                                                                                                                                                                                                                            | No input signal, 1:10 Probe                   | <b>16.384MHz ... 25.600MHz</b>                                                                                    | <b>24.88MHz</b>           | In range → <b>passed</b>                                             |                                                                                       |
| MCLK Frequency                                                                                                                                                                                                                                               | No input signal                               | <b>1.024MHz ... 25.600MHz</b>                                                                                     | <b>12.3MHz</b>            | In range → <b>passed</b>                                             |                                                                                       |
| LRCLK Frequency                                                                                                                                                                                                                                              | No input signal                               | <b>4kHz ... 50kHz</b>                                                                                             | <b>48kHz</b>              | In range → <b>passed</b>                                             |                                                                                       |
| SCLK Frequency                                                                                                                                                                                                                                               | No input signal                               | $LRCLK \times 64 = 256kHz \dots 3.2MHz$                                                                           | <b>3.07MHz</b>            | In range → <b>passed</b>                                             |                                                                                       |
| SDOUT signal present                                                                                                                                                                                                                                         | No input signal                               | <b>Signal present</b>                                                                                             | <b>Yes</b>                | As expected → <b>passed</b>                                          |                                                                                       |
| Output signal after Reconstruction filter, left channel                                                                                                                                                                                                      | Input signal:<br>1mVpp 1kHz sine              | No attenuation @ 1kHz → <b>500mV<sub>PP</sub></b>                                                                 | <b>533mV<sub>PP</sub></b> | Acceptable with tolerance → <b>passed</b>                            |                                                                                       |
| Output signal after Headphone Amplifier, left channel                                                                                                                                                                                                        | Input signal:<br>1mVpp 1kHz sine              | Amplification factor = 5 → <b>2.5V<sub>PP</sub></b>                                                               | <b>2.62V<sub>PP</sub></b> | Acceptable with tolerance → <b>passed</b>                            |                                                                                       |
| Output signal after Reconstruction filter, right channel                                                                                                                                                                                                     | Input signal:<br>1mVpp 1kHz sine              | No attenuation @ 1kHz → <b>500mV<sub>PP</sub></b>                                                                 | <b>536mV<sub>PP</sub></b> | Acceptable with tolerance → <b>passed</b>                            |                                                                                       |
| Output signal after Headphone Amplifier, right channel                                                                                                                                                                                                       | Input signal:<br>1mVpp 1kHz sine              | Amplification factor = 5 → <b>2.5V<sub>PP</sub></b>                                                               | <b>2.60V<sub>PP</sub></b> | Acceptable with tolerance → <b>passed</b>                            |                                                                                       |
| Frequency sweep                                                                                                                                                                                                                                              | Input signal: Frequency sweep 400Hz ... 20kHz | <b>No attenuation</b> in FFT spectrum                                                                             | <b>Yes</b>                | As expected → <b>passed</b>                                          |                                                                                       |

Information on the conducted tests:

**Date:** 31.10.2025, 07.11.2025 and 14.11.2025

**Location:** ZHAW, Technikumstrasse 71, 8400 Winterthur, Room TS04.11

**Test personnel:** Thomas Perri and Bogdans Grebnevs

**Equipment:** Keysight U3402A digital multimeter, Tektronix TDS 2012C oscilloscope, TTI EL302RT laboratory power supply

*For further detailed information see the accompanying Excel table.*

## 4. Test Analysis

During testing, several issues related to missing or incorrect electrical connections were identified and subsequently corrected.

### **Problem 1: Incorrect power-supply flag naming**

During schematic design, some power-supply flags were inconsistently named across different sheets. As a result, the corresponding nets were not connected in the PCB layout.

**Solution:** Two patch wires were added: one from the 3.3 V line to C29 and another from JP5 to R27.

After applying these modifications, the power-supply functioned correctly throughout the entire device.

### **Problem 2: Reference-voltage (VCOM) buffer of the input filter**

Similar to Problem 1, the signal flags for the VCOM reference voltage were mixed up. This caused an unintended bridging of the buffer stage and led to a short between the codec and filter VCOM pins.

**Solution:** The affected PCB traces were cut in three locations, and two patch wires were added: one from the codec's VCOM output (via) to the input pin of the buffer op-amp, and another from R37 to C37.

After implementing these changes, both the codec VCOM output and the buffer output showed correct levels, restoring proper operation of the input filter.

After implementing these corrections, all tests were completed successfully, with only minor deviations that stayed within acceptable limits. In summary, the DUT has passed the hardware validation and is ready to proceed to the next phase of firmware development.

